COMPETENCY STANDARD

FOR

ELECTRICAL INSTALLATION AND MAINTENANCE

(CONSTRUCTION SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kalyan Bhaban (Level-15)
71-72 Eskaton Garden, Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: https://sicip.gov.bd/

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity.
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 hrs.)

Code	**Unit of Competency**	**Elements of Competency**	**Duration (Hours)**
SICIP-CON-EIM-01-G	Apply occupational health and safety (OHS) practice in the workplace	1. Identify OHS policies and procedures. 2. Apply personal health and safety practices. 3. Report hazards and risks 4. Respond to emergencies.	10
SICIP-CON-EIM-02-G	Carry out workplace interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	10
SICIP-CON-EIM-03-G	Operate in a team environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	10
		Total Hour	**30 Hrs.**

Sector Specific Competencies (40 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-CON-EIM-1-S	Apply green practices in Construction Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
SICIP-CON-EIM-2-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	15
SICIP-CON-EIM-3-S	Carry Out Measurements and Calculations	1. Check usability of measuring devices 2. Carry out accurate construction work measurements 3. Execute simple construction work calculations 4. Clean and maintain measuring instruments	15
		Total Hour	**40 Hrs.**

Occupation Specific Competencies (290 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-CON-EIM-01-O	Perform Channel Wiring.	1. Interpret drawings and specifications. 2. Collect tools, equipment and materials 3. Draw the layout, set channels and cables 4. Install boards and set all other accessories of wiring 5. Perform circuit operation as per diagram and layout 6. Clean the work place	50
SICIP-CON-EIM-02-O	Perform Concealed Conduit Wiring	1. Plan and mark the layout for concealed wiring 2. Cut walls and floors for conduit installation 3. Install and secure conduits within walls and floors 4. Pull and install electrical wires through conduits 5. Install switches, sockets and fixtures into gang boxes 6. Test wiring installation for functionality and safety	50
SICIP-CON-EIM-03-O	Perform Service Connection	1. Interpreted drawings and specifications 2. Install cables for service connection 3. Install and dress electrical distribution board 4. Install energy meter	45
SICIP-CON-EIM-04-O	Install Earthing and Atmospheric Lightning Protection System	1. Identify the type of earthing to be used 2. Identify the type of lightning protection system to be used	35

Code	Unit of Competency	Elements of Competency	Duration (hours)
		3. Select and collect tools, equipment and materials 4. Excavate the hole for earthing element installation 5. Install earthing components 6. Finish earth pit chamber for pipe earthing method 7. Install lightning protection system 8. Clean and maintain the work area	
SICIP-CON-EIM-05-O	Perform Motor Connection	1. Identify and select controlling and protective devices for motor connection 2. Install, controlling and protective devices 3. Perform single phase motor connection 4. Perform three phase motor connection	40
SICIP-CON-HIM-06-O	Operate residential generator	1. Install and connect an Automatic Transfer Switch (ATS) and changeover switch 2. Connect the generator to the main electrical panel 3. Prepare generator for operation 4. Start and stop the generator 5. Perform post-operation checks and maintenance	35
SICIP-CON-HIM-07-O	Install and Troubleshoot Solar Electrical System	1. Estimate electrical load of customer 2. Identify tools, equipment, and materials 3. Set solar panel 4. Install solar home	35

Code	Unit of Competency	Elements of Competency	Duration (hours)
		system and accessories 5. Diagnose faults in solar home system unit and wiring 6. Repair faults in solar home system unit and wiring 7. Clean and store the tools and materials	
		Total Hours	**00 Hrs.**

COMPETENCY STANDARD: ELECTRICAL INSTALLATION AND MAINTENANCE

Unit of Competency:	Nominal Duration:	Unit Code:
APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	10 hrs.	SICIP-CON-EIM-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	1.4 OHS policies and procedures are interpreted in the workplace including **<u>personal protective equipment (PPE).</u>** 1.5 Common health issues are recognized. 1.6 Common safety issues are identified and applied.
3. Report hazards and risks	1.1 Hazards and risks are identified. 1.2 Hazards and risks assessment and controls are interpreted 1.3 Hazards and risk assessment are reported
4. Respond to emergencies	1.1 Respond to alarms and warning devices. 1.2 **<u>Emergency response plans and procedures</u>** are responded to. 1.3 **<u>First aid procedures</u>** during emergency situations are identified.

Range of variables:

Variables	**Range** (may include but not limited to)
1. OHS policies	1.1 Organizational OHS polices 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Personal protective equipment	2.1 Safety glasses 2.2 Ear plugs 2.3 Gloves 2.4 Apron

	2.5 Helmet 2.6 Mask 2.7 Safety shoes
3. Emergency response plans and procedures	3.1 Firefighting procedures 3.2 Earthquake response procedures 3.3 Emergency response plans and procedures 3.4 Medical and first aid
4. First aid procedure	4.1 Washing of open wound 4.2 Washing chemically infected area 4.3 Applying bandage 4.4 Taking appropriate medicine

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	1.8 Identifying OHS policies and procedures 1.9 Applying personal health and safety practices 1.10 Reporting hazards and risks 1.11 Responding to emergencies
3. Underpinning Attitudes	1.12 Commitment to occupational health and safety 1.13 Promptness in carrying out activities 1.14 Sincere and honest to duties 1.15 Environmental concerns 1.16 Eagerness to learn 1.17 Tidiness and timeliness 1.18 Respect for rights of peers and seniors in the workplace 1.19 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.20 Workplace (simulated or actual) 1.21 Workplace documents, signs and symbols 1.22 Codes of conduct 1.23 Projector 1.24 Learning manual 1.25 Tools, equipment and facilities appropriate to the process or activities. 1.26 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified OHS policies and procedures 1.2 Applied personal health and safety practices (including PPE) 1.3 Reported hazards and risks 1.4 Responded to emergencies.
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-EIM-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **courteous manner** to gather and convey information. 1.4 **Workplace procedures and matters** are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from **appropriate sources**. 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	1.1 Notes 1.2 Arranging a meeting 1.3 Agenda 1.4 Simple reports such as progress and incident reports 1.5 Job sheets 1.6 Operational manuals 1.7 Brochures and promotional material 1.8 Visual and graphic materials 1.9 Standards 1.10 OHS information 1.11 Signs
3. Appropriate sources	1.1 Human Resources (HR) Department 1.2 Managers 1.3 Supervisors 1.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	1.5 Demonstrating workplace communication and etiquette 1.6 Interpreting workplace instructions and symbols 1.7 Demonstrating active participation in workplace meeting 1.8 Applying professional ethics at work
3. Underpinning Attitudes	1.9 Commitment to occupational health and safety 1.10 Promptness in carrying out activities 1.11 Sincere and honest to duties 1.12 Environmental concerns 1.13 Eagerness to learn 1.14 Tidiness and timeliness 1.15 Respect for rights of peers and seniors in the workplace 1.16 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process

	or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-CON-EIM-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through **sharing information** or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	1.1. Identifying own role and responsibilities within team 1.2. Communicating and co-operating with team members 1.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	1.4. Commitment to occupational health and safety 1.5. Promptness in carrying out activities 1.6. Sincere and honest to duties 1.7. Eagerness to learn 1.8. Tidiness and timeliness 1.9. Environmental concerns 1.10. Respect for rights of peers and seniors at workplace 1.11. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 1.12. Workplace (simulated or actual) 1.13. Workplace documents, Projector 1.14. Learning manual 1.15. Tools, equipment and facilities appropriate to the process or activities. 1.16. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test

	2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

<table>
<tr><td>Unit of Competency:

APPLY GREEN PRACTICES IN CONSTRUCTION SECTOR</td><td>Nominal Duration:
10 hrs.</td><td>Unit Code:
SICIP-CON-EIM-01-S</td></tr>
<tr><td colspan="3">Unit Descriptor:

This unit covers the skills, knowledge and attitudes required for apply green practices in construction sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.</td></tr>
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Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Green concepts in the construction sector</u>** are understood. 1.2 **<u>Sustainable building materials</u>** are recognized. 1.3 Energy-efficient designs are interpreted. 1.4 **<u>Water conservation practices</u>** are understood. 1.5 Construction waste management is recognized. 1.6 **<u>Building codes and regulations</u>** are interpreted. 1.7 **<u>Green building certifications</u>** are understood.
2. Minimize resources use in the workplace	2.1 Resource consumption in the construction sector is regularly monitored. 2.2 **<u>Energy-efficient equipment and technologies</u>** are utilized. 2.3 Material selection is optimized. 2.4 Water usage is managed by using water-saving systems. 2.5 Optimized building designs are implemented. 2.6 Sustainable practices are followed.
3. Implement waste management practices	3.1 **<u>Types of waste</u>** generated in the construction sector. 3.2 Waste segregation is implemented on-site. 3.3 Recycling programs are recognized. 3.4 On-site storage and disposal methods are described. 3.5 Construction processes are followed ensuring waste management practices.

Range of Variables

Variable	Range May include but not limited to:
1. Green concepts in the construction sector	1.1 Sustainable Building Materials 1.2 Energy-Efficient Design 1.3 Water Conservation Measures 1.4 Waste Management and Recycling 1.5 Site Environmental Protection

	1.6 Renewable Energy Integration
2. Sustainable building materials	2.1 Recycled and Reclaimed Materials 2.2 Renewable Natural Materials 2.3 Low-Emission and Non-Toxic Materials 2.4 High-Performance Insulation Materials 2.5 Durable and Long-Lasting Materials 2.6 Innovative Eco-Friendly Materials
3. Water conservation practices	3.1 Rainwater Harvesting Systems 3.2 Greywater Recycling 3.3 Low-Flow Fixtures and Fittings 3.4 Efficient Site Water Management 3.5 Landscaping with Drought-Resistant Plants 3.6 Water Leak Detection and Repair Systems 3.7 Use of Water-Efficient Appliances and Equipment
4. Building codes and regulations	4.1 Bangladesh National Building Code (BNBC) 4.2 Environmental Conservation Rules (1997) 4.3 Water Supply and Sewerage Authority (WASA) Regulations 4.4 Fire Service and Civil Defense Regulations 4.5 Bangladesh Labour Act (2006) and Construction Safety 4.6 Local Government (City Corporation and Municipal) Building Bye-laws 4.7 Energy Efficiency and Green Building Guidelines
5. Green building certifications	5.1 LEED (Leadership in Energy and Environmental Design) 5.2 BREEAM (Building Research Establishment Environmental Assessment Method) 5.3 WELL Building Standard 5.4 EDGE (Excellence in Design for Greater Efficiencies) 5.5 Green Globe Certification 5.6 Green Rating for Integrated Habitat Assessment (GRIHA)
6. Energy-efficient equipment and technologies	6.1 High-Efficiency HVAC Systems 6.2 LED and Smart Lighting Systems 6.3 Energy-Efficient Construction Machinery 6.4 Building Automation and Control Systems (BACS) 6.5 Solar-Powered Equipment and Technologies 6.6 Insulation and Building Envelope Technologies
7. Types of waste	7.1 Concrete and Masonry Waste 7.2 Wood Waste 7.3 Metal Waste 7.4 Plastic and Packaging Waste 7.5 Hazardous Waste 7.6 Soil and Excavation Waste 7.7 Glass and Ceramics Waste

Curricular Content Guide

1. Underpinning Knowledge	1.1 Fundamental principles of green construction. 1.2 Benefits, environmental footprint, and lifecycle considerations 1.3 Overview of sustainable building materials 1.4 Relevant building codes and environmental regulations 1.5 Common types of construction waste 1.6 Waste segregation principles and methods 1.7 Green building certifications 1.8 Principles of integrated design and project management 1.9 Sustainable construction techniques
2. Underpinning Skills	2.1 Identifying and selecting sustainable building materials 2.2 Interpreting and applying energy-efficient design principle 2.3 Establishing and managing on-site waste storage systems 2.4 Segregating construction waste on-site into recyclable, reusable, and hazardous categories 2.5 Coordinating with recycling facilities and waste disposal contractors 2.6 Following building codes, environmental regulations, and green certification 2.7 Documenting waste generation and disposal activities
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace

	1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **USE HAND AND POWER TOOLS**	**Nominal Duration:** 15 hrs.	**Unit Code:** SICIP-CON-EIM-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools for usability, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1 Appropriate **<u>hand tools</u>** are selected. 2.2 Safety precautions are ensured before using hand tools. 2.3 Unsafe or faulty hand tools are identified and marked for repair. 2.4 **<u>Measuring tools</u>** are checked and calibrated before use. 2.5 Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate **<u>power tools</u>** are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after

	use and reported. 4.3. Appropriate lubricant is applied after use and prior to storage. 4.5. Defective hand and power tools are inspected and repaired or replaced. 4.6. Hand and power tools are stored and secured in accordance with workplace requirements.

Range of Variables

Variable	**Range** May include but not limited to:
1. Hand tools	Hammer 1.1. Bench vice 1.2. Files 1.3. Punches 1.4. Chisels 1.5. Wrenches 1.6. Pliers 1.7. Scriber 1.8. Scraper 1.9. Screwdrivers 1.10. Dividers 1.11. Surface plate 1.12. Gauge 1.13. Tap sets 1.14. Die sets 1.15. Hacksaw 1.16. Socket spanners 1.17. Spanners 1.18. Vice grip 1.19. Wire cutters 1.20. Drills 1.21. Drill buts 1.22. Grinder 1.23. Clamps 1.24. Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine

	2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Testers 3.3 Megger 3.4 Measuring tape 3.5 Hose level 3.6 Water level 3.7 Calipers 3.8 Steel rule 3.9 Meter rule 3.10 Spirit level 3.11 Protractor 3.12 Tri-square 3.13 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on types of hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual)

	4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **CARRY OUT MEASUREMENTS AND CALCULATIONS**	**Nominal Duration:** 15 hrs.	**Unit Code:** SICIP-CON-EIM-03-S

Unit Descriptor:

This unit covers the skills, knowledge and attitudes required to carry-out measurements and calculations. It specifically includes the task of checking usability of measuring devices, carrying out accurate construction work measurements, executing simple construction work calculations and cleaning and maintaining measuring instruments.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check usability of measuring devices	1.1 Appropriate **measuring device** is selected for the job. 1.2 Applications of measuring device is determined. 1.3 Usability of measuring device is checked and verified. 1.4 Measuring device is prepared.
2. Carry out accurate construction work measurements	2.1 Measurements are obtained using appropriate measuring device. 2.2 **Systems of measurements** are identified and converted where necessary. 2.3 Results are confirmed and recorded.
3. Execute simple construction work calculations	3.1 Simple calculations involving **four basic mathematical operations** are executed. 3.2 Other operations are used to complete tasks in construction works 3.3 Appropriate formulas for calculating quantities of materials are selected. 3.4 Calculations are performed and verified. 3.5 Material quantities are calculated. 3.6 Results are interpreted and communicated to authority.
4. Clean and maintain measuring instruments	4.1 Dust and foreign matters are removed from measuring instrument 4.2 Condition of instrument is checked 4.3 Appropriate lubricant is applied after use and prior to storage 4.4 Measuring instruments are checked and calibrated. 4.5 Instrument is stored in accordance to workplace procedure

Range of Variables

Variable	Range May include but not limited to:

1. Measuring device	1.1 Slide calipers 1.2 Steel tape measure 1.3 Steel rule 1.4 Square 1.5 Tri-square 1.6 Feeler gauges 1.7 Water level 1.8 Hose level 1.9 Thermometers 1.10 Protractor
2. Systems of measurements	2.1 ISO standard 2.2 English system 2.3 Metric system
3. Four basic mathematical operations	3.1 Addition 3.2 Subtraction 3.3 Multiplication 3.4 Division

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and principles of operation of measuring devices 1.2 The ISO standard of measurements 1.3 Methods of measurement and calculation 1.4 Fraction and decimals 1.5 Linear measurement 1.6 Units of conversion and conversion factors in measurements 1.7 Dimensioning and fits and tolerances 1.8 Calculating ratio and proportion 1.9 Care in the use of measuring devices
2. Underpinning Skills	2.1 Selecting appropriate measuring device for the job 2.2 Checking and verifying usability of measuring device 2.3 Obtaining measurements using appropriate measuring device. 2.4 Confirming measurements and recording results 2.5 Carrying out simple calculations involving four basic mathematical operations 2.6 Calculating material quantities 2.7 Interpreting and communicating results to authority 2.8 Cleaning and storing measuring instruments
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the

	workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Projector 4.4 Learning manual 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Selected appropriate measuring device for the job. 1.2 Checked and verified usability of measuring device. 1.3 Obtained measurements using appropriate measuring device. 1.4 Confirmed measurements and recorded results. 1.5 Carried out Simple calculations involving four basic mathematical operations. 1.6 Calculated material quantities. 1.7 Interpreted and communicated Results to authority.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: **PERFORM CHANNEL WIRING**	Nominal Duration: 50 Hrs.	Unit Code: SICIP-CON-EIM-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform channel wiring. It specifically includes the task of interpreting drawings and specifications, collecting tools, equipment and materials, drawing the layout, set channels and cables, installing boards and set all other accessories of wiring, performing circuit operation as per diagram and layout and cleaning the workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret drawings and specifications	1.1. Drawings are collected and interpreted. 1.2. Sign and symbols are identified. 1.3. Terms and abbreviations are identified. 1.4. Specifications are interpreted.
2. Collect tools, equipment and materials	2.1 **Hand Tools, power tools, equipment** and **materials** are collected. 2.2 Tools, equipment and materials are checked for usability.
3. Draw the layout, set channels and cables	3.1 **PPE** is collected and used. 3.2 Wiring layout is drawn according to supplied drawing. 3.3 Rowel plug points are located, drilled and inserted as per procedure. 3.4 Bottom part of the channels are installed and screwed. 3.5 Cables with ECC are laid on the bottom part of the channel
4. Install boards and set all other accessories of wiring	4.1 **Boards** are collected and fitted as per wiring diagram. 4.2 Switches, sockets, fan regulator and Ballast are fitted on the board with screw 4.3 Switches, sockets and fan regulator are connected to the circuits 4.4 Ceiling rose and different types of holders are fitted on the board 4.5 Fixtures are connected to the circuit 4.6 **Fuse and Circuit Breakers** are connected and fitted on the board.
5. Perform circuit operation as per diagram and layout	5.1 Bottom parts of the channels are placed and set according to drawing on the board 5.2 Cables are drawn through the bottom part of the channels

	5.3 Circuit materials required for the specified circuit are placed on the board 5.4 Other accessories are connected and fitted 5.5 The bottom parts of the channels are covered with upper part of the channel
6. Clean the workplace	6.1 Tools and equipment are prepared for cleaning 6.2 Tools and equipment are stored as per standard 6.3 West materials are disposed as per workplace standard

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Hand Tools	1.1 Adjustable wrench 1.2 Wire stripper 1.3 Mallet 1.4 C-clamp 1.5 Chisels: (a) Wooden, (b) Cold 1.6 Drill bits 1.7 Files: (a) Flat (b) Round (c) Half round 1.8 Hacksaw 1.9 Hammers: (a) Ball peen, (b) Claw 1.10 Hand drill 1.11 Measuring tape 1.12 Pliers: (a) Combination pliers, (b) Cutting pliers, (c) Diagonal cutting pliers, (d) Long nose pliers, 1.13 Punches 1.14 Screwdrivers: (a) Star, (b) Flat, (c)Connecting 1.15 Try square 1.16 Neon tester 1.17 Wire cutters 1.18 S.W.G. 1.19 Set squares 1.20 Electrician knife 1.21 Ladder
2. Power Tools	2.1 Electric drill machine 2.2 Grinders 2.3 Soldering iron
3. Equipment	3.1 Multi meter / AVO meter 3.2 Earth tester 3.3 Digital weight machine
4 Materials	4.1 Refers to all electrical materials included but not limited to the following: 4.2 Channel (1/2”, ¾”, 1”,1.25”, 1.5” PVC) 4.3 GI Wire 4.4 Elbow

	4.5 Bend 4.6 PVC circular box 4.7 Rowel plug 4.8 Saddle 4.9 Screw 4.10 Cable lugs 4.11 Cable tie 4.12 Thread ball 4.13 Insulating clip 4.14 Flexible conduit 4.15 Plastic forma 4.16 Electric soldering lead 4.17 Plastic tape 4.18 Cable (PVC, VIR)
5 PPE	5.1 Uniform 5.2 Hand Gloves 5.3 Helmet 5.4 Goggles 5.5 Safety shoes
6 Boards	6.1 Plastic board 6.2 Ebonite boards 6.3 Wooden boards
7 Fuses & Circuit Breakers	7.1 Rewire cable fuse 7.2 Cartridge fuse 7.3 Glass fuse 7.4 HRC fuse 7.5 Single pole MCB 7.6 Double pole MCB 7.7 MCCB 7.8 Earth leakage circuit breaker (ELCB)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Drawings and specifications. 1.2 Signs and symbol identification 1.3 Terms and abbreviations identification 1.4 Drawings interpretation. 1.5 Different type of fittings and fixtures 1.6 Quality of fittings and fixtures use for installing 1.7 Different types of tools equipment and machinery used for installing fittings and fixtures 1.8 Fittings and fixture installation
2. Underpinning Skills	2.1 Using PPE 2.2 Selecting appropriate tools and equipment. 2.3 Selecting appropriate materials. 2.4 Checking specifications 2.5 Locating points 2.6 Installing channel

	2.7 Laying cables 2.8 Selecting appropriate materials as per schedule 2.9 Fitting all switches, sockets, and fixtures 2.10 Installing fittings and fixtures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Tidiness and timeliness 3.6 Concerned for proper use of tools
4 Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Materials 4.4 Projector 4.5 Stationary 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Interpreted drawings and specifications 1.2 Drew the layout, set channels, and cables 1.3 Installed boards and set all other accessories of wiring 1.4 Performed circuit connection as per diagram and layout
2. Methods of Assessment	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM CONCEALED CONDUIT WIRING**	**Nominal Duration:** 50 Hrs**.**	**Unit Code:** SICIP-CON-EIM-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform concealed conduit wiring. It specifically includes the task of planning and marking the layout for concealed wiring, cutting walls and floors for conduit installation, installing and securing conduits within walls and floors, pulling and installing electrical wires through conduits, installing switches, sockets and fixtures into gang boxes and testing wiring installation for functionality and safety.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Plan and mark the layout for concealed wiring	1.1. **<u>Hand Tools, power tools, equipment and materials</u>** are collected. 1.2. **<u>PPE</u>** is collected and used. 1.1. Wiring paths are planned to minimize interference with other building services. 1.2. Layout markings are made accurately on walls, ceilings, and floors as per design specifications. 1.3. **<u>Boards</u>**, Junction box, outlet positions and ceiling hooks are marked based on accessibility and usage requirements. 1.4. Measurements are taken to ensure alignment with electrical drawings. 1.5. Safety precautions are taken to avoid structural damage during installation.
2. Cut walls and floors for conduit installation	2.1 **<u>Groove cutting tools</u>** are selected depending on the types of materials of the wall or floor. 2.2 Conduit channels are cut with precision to avoid damage to structural components. 2.3 Grooves for **<u>mounting boxes</u>** are cut to the specified dimensions and positions. 2.4 Ceiling hooks are positioned according to the layout. 2.5 Depth and width of the channels are measured for proper conduit placement.
3. Install and secure conduits within walls and floors	3.1 Conduits are placed into pre-cut channels according to the marked layout. 3.2 Conduits are securely fastened using appropriate **<u>fixing methods</u>**. 3.3 Bends, joints, and fittings are installed to avoid obstructions or sharp angles. 3.4 Conduits are inspected to ensure no blockages or damage before wiring.

4. Pull and install electrical wires through conduits	4.1 Wires are selected based on the circuit's load requirements and insulation ratings. 4.2 Wire pulling is conducted using fish wire preventing insulation damage. 4.3 Wiring is installed in a manner that allows easy future maintenance and upgrades. 4.4 Identification tags or color coding is used to distinguish different circuits. 4.5 Wiring paths are checked for continuity and secure connections.
5. Install switches, sockets and fixtures into gang boxes	5.1 Switches and sockets are mounted level with the wall surface for aesthetic finish. 5.2 Electrical connections are made according to wiring diagrams and circuit plans. 5.3 Fixtures are securely mounted into gang boxes and aligned for functionality. 5.4 Protective covers and faceplates are installed for safety and appearance.
6. Test wiring installation for functionality and safety	6.1 Continuity tests are performed to verify all connections are properly made. 6.2 Insulation resistance tests are conducted to ensure no leakage currents. 6.3 Polarity and grounding tests are completed to ensure system safety. 6.4 Any faults identified during testing are rectified.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Hand Tools	1.1 Adjustable wrench 1.2 Wire stripper 1.3 Bolt cutters 1.4 Mallet 1.5 C-clamp 1.6 Chisels: (a) Wooden, (b) Cold 1.7 Drill bits 1.8 Files: (a) Flat,(b) Round,(c) Half round 1.9 Hacksaw 1.10 Hammers: (a) Ball peen, (b) Claw 1.11 Hand drill 1.12 Measuring Tapes 1.13 Paint Brushes/Rollers 1.14 Pliers: (a) Combination Pliers, (b) Cutting Pliers, (c) Diagonal cutting Pliers, (d) Long Nose Pliers, 1.15 Punches 1.16 Screwdrivers: (a) Star, (b) Flat, (c) Connecting 1.17 Try square

	1.18 Neon tester 1.19 Wire cutters 1.20 S.W.G. 1.21 Set squares 1.22 Electrician knife 1.23 Ladder
2. Power Tools	2.1 Electric drill machine 2.2 Grinders 2.3 Soldering iron
3. Equipment	3.1 Meggar 3.2 Calculator 3.3 Multi meter/AVO meter 3.4 Earth tester 3.5 Digital weight machine
4. Materials	4.1 Refers to all construction(electrical) materials included but not limited to the following: 4.2 Conduit PVC 4.3 GI Wire 4.4 Elbow 4.5 Bend 4.6 PVC circular box 4.7 Rowel plug 4.8 Saddle 4.9 Screw 4.10 Cable lugs 4.11 Cable tie 4.12 Thread ball 4.13 Insulating clip 4.14 Flexible conduit 4.15 Plastic forma 4.16 Electric soldering lead 4.17 Plastic tape 4.18 Cable
5. PPE	5.1 Hand gloves 5.2 Helmet 5.3 Goggles 5.4 Safety shoes
6. Boards	6.1 Plastic board 6.2 Ebonite boards 6.3 Metal Box
7. Mounting boxes	7.1 Gang boxes 7.2 Outlet boxes 7.3 Junction boxes
8. Fixing methods	8.1 Screw and anchor 8.2 Strap or ties 8.3 Conduit insert 8.4 Conduit brackets

	8.5 Adhesive mounts

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic principles of electrical design, circuits, and wiring regulations. 1.2 Knowledge of electrical safety standards and practices. 1.3 Understanding of building structures, wall and floor materials. 1.4 Awareness of other building services to avoid interference. 1.5 Proper use and maintenance of hand tools, power tools, and groove-cutting tools. 1.6 Selection of appropriate tools based on materials (e.g., wood, concrete). 1.7 Importance and usage of Personal Protective Equipment (PPE). 1.8 Testing methods: continuity, insulation resistance, polarity, and grounding. 1.9 Types of conduits, wires, switches, sockets, and fixtures. 1.10 Standards and compatibility of electrical components.
2. Underpinning Skills	2.1 Marking conduit paths and electrical layouts accurately on walls, ceilings, and floors. 2.2 Cutting walls and floors with groove-cutting tools for conduit installation without damaging structures. 2.3 Installing conduits in pre-cut channels and securely fastening them using appropriate methods. 2.4 Pulling wires through conduits without damaging insulation or causing obstructions. 2.5 Measuring conduit channels, depths, and widths precisely to ensure proper alignment with drawings. 2.6 Testing wiring systems for continuity, insulation resistance, polarity, and grounding. 2.7 Identifying and rectifying wiring faults during testing and installation processes. 2.8 Interpreting electrical plans, wiring diagrams, and technical specifications accurately. 2.9 Securing switches, sockets, and fixtures properly within gang boxes for functionality and safety. 2.10 Using fish wire and other tools efficiently to guide wiring through conduits.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness

	3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Materials 4.4 Projector 4.5 Stationary 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Ensuring accurate layout and installation of conduits in alignment with electrical plans. 1.2 Adhering to safety standards, using PPE, and preventing structural damage during installation. 1.3 Performing functional and safety tests, such as continuity and insulation resistance. 1.4 Identifying and rectifying faults to ensure proper system functionality before completing the installation. 1.5 Maintaining proper installation of switches, sockets, and fixtures for both functionality and aesthetics.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM SERVICE CONNECTION**	**Nominal Duration:** 45 Hrs**.**	**Unit Code:** SICIP-CON-EIM-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform service connection. It specifically includes the tasks of interpreting drawings and specifications, installing cables for service connection, installing and dress electrical distribution board and installing energy meter.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret drawings and specifications	1.1 **Drawing** is collected and interpreted. 1.2 **Sign and symbols** are identified. 1.3 Terms and abbreviations are identified. 1.4 **Specifications** are interpreted.
2. Install cables for service connection	2.1 Distance between distribution pole and meter is checked and measured. 2.2 Distance between main switch and meter is checked and measured. 2.3 Size of cables are selected as per load 2.4 Quality cables are selected and collected for service connection 2.5 Collected cables are cut and set. 2.6 Cables are held on and clamped properly with distribution pole. 2.7 Cables are joined and connected with the pole and energy meter.
3. Install and dress electrical distribution board	3.1 The installation site is assessed to determine the appropriate size and type of distribution board based on connected devices. 3.2 **Circuit breakers,** bus bars, and isolators are securely installed as per manufacturer guidelines. 3.3 **Cable termination** is completed with cable lugs to securely connect cables to the bus bar, circuit breakers, and neutral/earth links. 3.4 Indicator lamps are fitted and tested to show the operational status of the distribution board. 3.5 Each circuit is labeled for easy identification and maintenance. 3.6 Power supply connections, including neutral and earth wires, are made correctly, with neutral connected to the neutral link and earth to the earth link.

	3.7 The distribution board is securely mounted, and cables are neatly dressed with ties for organization.
4. Install energy meter	4.1 The correct energy meter is selected based on load requirements and specifications. 4.2 The mounting position is prepared and energy meter is securely mounted as per manufacturer guidelines. 4.3 Service lines (incoming supply) are identified for proper connection. 4.4 Cable insulation is stripped, and lugs are crimped for connection. 4.5 Service line connections and earth connection to the energy meter are securely tightened. 4.6 Energy meter is connected to the main switch and connections are tightened. 4.7 Connections are tested to ensure proper functionality and safety.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Drawings	1.1 Sketch 1.2 Blue Print 1.3 Electrical drawings
2. Specification's	2.1 Product specification's 2.2 Performance specification's 2.3 Method specifications 2.4 Specification manuals
3. Sign and symbols	3.1 Include all signs and symbol associated with Electrical and maintenance work 3.2 Drawing symbol 3.3 Connection symbols 3.4 Load symbols 3.5 Socket symbols 3.6 Main switch symbols 3.7 Supply symbols 3.8 Danger symbols 3.9 Switch board symbols 3.10 Conduit board 3.11 Circuit Breaker symbol 3.12 Protective device symbol
4. Circuit breakers	4.1 According to pole 4.2 Single pole 4.3 Double pole 4.4 Triple pole 4.5 Four pole 4.6 According to types

	4.7 Miniature Circuit Breaker (MCB) 4.8 Moulded Case Circuit Breakers (MCCB) 4.9 Residual Current Circuit Breaker (RCCB) 4.10 Residual Current Device (RCD) 4.11 Earth Leakage Circuit Breaker (ELCB) 4.12 Earth Leakage Circuit Breaker Relay (ELR)
5. Cable termination	5.1 Cable/Wire preparation 5.2 Measurements and Marking 5.3 Cutting 5.4 Stripping 5.5 Scratching 5.6 Selecting the correct size of cable Lug or Ferrule. 5.7 Crimped (Crimping Tools & Hydraulic crimping Tools) 5.8 Insulate with Heat Shrink 5.9 Screw-on (Connectors / Terminate / Termination Block / Busbar)

Curricular Content Guide

1. Underpinning Knowledge	1.1. Electrical drawings, signs, and symbols (e.g., main switch symbols, circuit breaker symbols, load symbols). 1.2. Types and selection of cables based on load requirements. 1.3. Methods of interpreting electrical specifications, drawings, and manuals. 1.4. Different types of circuit breakers (e.g., MCB, MCCB, RCCB, ELCB) and their applications. 1.5. Procedures for measuring and checking distances for service connections. 1.6. Installation techniques for electrical distribution boards, circuit breakers, bus bars, and energy meters. 1.7. Safety standards and regulations related to electrical service connections. 1.8. Correct cable termination techniques, including lug crimping and insulation. 1.9. Principles of power supply connections, including neutral and earth connections. 1.10. Maintenance and troubleshooting of installed systems, such as energy meters and distribution boards.
2. Underpinning Skills	2.1 Interpret electrical drawings and specifications accurately. 2.2 Select appropriate cables and materials based on load and specifications. 2.3 Measure distances between distribution poles, meters, and main switches.

	2.4 Install and securely fasten cables on distribution poles and energy meters. 2.5 Join and connect cables properly to poles and meters. 2.6 Install electrical distribution boards, ensuring components like circuit breakers and bus bars are securely placed. 2.7 Label circuits and ensure correct connections in distribution boards. 2.8 Mount and secure energy meters based on manufacturer guidelines. 2.9 Terminate cables with proper lugs and ensure insulation. 2.10 Test connections and equipment to ensure safe and functional installations. 2.11 Apply safety standards during all installation processes.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Tidiness and timeliness 3.6 Concerned for proper use of tools
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationery 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Assessment required evidence that the candidate: 1.2 Installed service cable, energy meter and main switch as per drawing 1.3 Connected energy meter and main switch as per drawing 1.4 Ability to interpret and apply electrical drawings, symbols, and specifications. 1.5 Correct selection and installation of cables, ensuring adherence to load and distance requirements. 1.6 Proper installation and connection of distribution boards and energy meters. 1.7 Accuracy in cable termination, ensuring secure and

	safe connections. 1.8 Consistent application of safety procedures and standards throughout the installation process. 1.9 Ability to troubleshoot and maintain installed electrical systems.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INSTALL EARTHING AND ATMOSPHERIC LIGHTNING PROTECTION SYSTEM**	**Nominal Duration:** 35 Hrs.	**Unit Code:** SICIP-CON-EIM-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install earthing and atmospheric lighting protection system. It specifically includes the tasks of identifying the type of earthing to be used, identifying the type of lightning protection system to be used, selecting and collecting tools, equipment and materials, excavating the hole for earthing element installation, installing earthing components, finishing earth pit chamber for pipe earthing method, installing lightning protection system and cleaning and maintaining the work area.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the type of earthing to be used	1.1 Types and **<u>method of earthing</u>** is identified in accordance to electrical plan / design 1.2 Types and sizes of earthing materials are identified in accordance to electrical plan / design
2. Identify the type of lightning protection system to be used	1.3 **<u>Types of lightning protection system</u>** is identified in accordance to electrical plan / design 1.4 Types and sizes of **<u>lightning protection system materials</u>** are identified in accordance to electrical plan / design
3. Select and collect tools, equipment and materials	1.5 **<u>Tools and equipment</u>** are selected and collected 1.6 Tools and equipment are checked for usability 1.7 Earthing materials are collected and checked for conformance in accordance to specification 1.8 Lightning protection materials are collected and checked for conformance in accordance to specification
4. Excavate the hole for earthing element installation	1.9 **<u>PPE</u>** is collected and used in accordance to OHS requirements 1.10 Hole is dug following with safety requirements 1.11 Hole is shaped and sized in accordance to electrical plan / design specification
5. Install earthing components	1.12 **<u>Earthing element</u>** is fitted in the bottom of the excavated hole following standard earthing procedure 1.13 Earth lead is connected to the earth element tightly and brought up the meter board through the conduit. 1.14 Powdered charcoal and salt are laid around the earthing element in accordance to workplace procedure 1.15 A proper sized and length of GI pipe is fitted from top of the earth element to the bottom of the earth pit chamber 1.16 Rest of the excavated hole is filled with earth

6. Finish earth pit chamber for pipe earthing method	1.17 Earth pit chamber is constructed with brick chips, cements sand and water mixture in accordance with standard/specification. 1.18 Pit chamber cover is made with G.I sheet in accordance with electrical plan/design 1.19 Pit cover is fitted/installed on the pit chamber 1.20 Check earth resistance in accordance with electrical plan / specification
7. Install lightning protection system	1.21 Lightning rod is installed at specified location 1.22 Earth down conductor is connected as per diagram 1.23 Performance of lightning protection system (LPS) is tested as per standard
8. Clean and maintain the work area	1.24 Electrical tools / instruments are cleaned and checked for operability 1.25 Work area is cleaned and waste materials are disposed in accordance with workplace requirements

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Earthing materials	1.1 Continuity conductor / cable 1.2 Earthing lead 1.3 Earth electrode / plate 1.4 Connector 1.5 G.I. pipe / conduit 1.6 Bolts and nuts 1.7 Powdered charcoal 1.8 Salt
2. Method of earthing	2.1 Pipe earthing 2.2 Rod earthing 2.3 Plate earthing 2.4 Waterman earthing 2.5 Strip or wire earthing
3. Tools and equipment	3.1 Tools 3.1.1 Measuring tape (30m) 3.1.2 Tri- square 3.1.3 Pocket tape (3m) 3.1.4 Claw hammer / crow bar 3.1.5 Wire stripper 3.1.6 Adjustable wrench 3.1.7 Bolt cutters 3.1.8 C-clamp 3.1.9 Chisels: (wooden, cold) 3.1.10 Drill bits 3.1.11 Files: (flat, round, half round) 3.1.12 Hand hacksaw 3.1.13 Hammers: (ball peen, claw) 3.1.14 Pliers: (combination pliers, cutting pliers, diagonal

	cutting pliers, long nose pliers) 3.1.15 Punches 3.1.16 Screwdrivers: (star, negative, positive) 3.1.17 Electrician knife 3.2 Equipment 3.2.1 Electric drill machine 3.2.2 Soldering iron 3.2.3 Megger tester 3.2.4 Multi meter/AVO meter 3.2.5 Earth Tester
4. PPE	4.1 Safety shoes 4.2 Safety gloves 4.3 Safety helmet 4.4 Uniform
5. Types of lightning protection system	5.1 Conventional LPS 5.2 Non-conventional LPS
6. Lightning protection system materials	6.1 Lightning rod (spike arrester) 6.2 Earth down conductor (arrester) 6.3 Check terminal 6.4 Earth led

Curricular Content Guide

1. Underpinning Knowledge	1.1. Types of earthing 1.2. Materials required for earthing 1.3. Procedure of earthing 1.4. Testing procedure of earthing 1.5. Types of LPS 1.6. Materials required for LPS 1.7. Installation procedure of LPS 1.8. Performance testing procedure of LPS
2. Underpinning Skills	2.1 Collecting earthing and LPS materials and checked for conformance in accordance to specification 2.2 Excavating a hole for the earthing element 2.3 Fitting earthing element in the bottom of the excavated hole following standard earthing procedure 2.4 Connecting earth lead to the earth element tightly and brought up the meter board through the conduit 2.5 Installing lightning rod at specified location 2.6 Connecting earth down conductor as per diagram 2.7 Testing the performance of lightning protection system (LPS) as per standard 2.8 Checking earth loop resistance using appropriate test instrument
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns

	3.5 Tidiness and timeliness 3.6 Concerned for proper use of tools
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Tools and equipment 4.4 Materials 4.5 Projector 4.6 Stationary 4.7 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Tested earth resistance as per standard procedure 1.2 Tested the performance of LPS as per standard procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM MOTOR CONNECTION**	Nominal Duration: 40 Hrs.	Unit Code: SICIP-CON-EIM-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform motor connection. It specifically includes the tasks of identifying and selecting controlling and protective devices for motor connection, Installing, controlling and protective devices, performing single phase motor connection and performing three phase motor connection.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and select controlling and protective devices for motor connection	1.1 **Manuals** and documents of controlling and protective devices are collected. 1.2 **Drawings** and **symbols** of controlling and protective devices are sorted. 1.3 Types of **controlling** and **protective devices** are listed
2. Install controlling and protective devices	2.1 Controlling and protective devices are selected and collected according to the need of the operations 2.2 Controlling and protective devices are installed according to the layout plan 2.3 Controlling and protective devices are set and connected to the motor
3. Perform single phase motor connection	3.1 The **motor** rated voltage, current, and power specifications are checked before installation. 3.2 The correct type and size of cable are selected according to motor load and circuit requirements. 3.3 Insulation is stripped, and cable lugs are crimped securely for motor and supply connections. 3.4 The live, neutral, and earth wires are connected to the motor's terminal block according to the wiring diagram. 3.5 Motor control switch is connected as per safety standards. 3.6 All connections are tightened, and the motor is tested for proper operation. 3.7 The motor's rotation direction is verified and adjusted if necessary to ensure correct functionality.
4. Perform three phase motor connection	4.1 The motor's rated voltage, current, and power specifications are verified before installation. 4.2 The correct type and size of cable are selected based on motor load and circuit requirements for both star and delta connections. 4.3 Insulation is stripped, and cable lugs are crimped securely for the motor and supply connections. 4.4 For star connection, the three phase wires are

	connected to the motor terminals as per the star configuration diagram. 4.5 For delta connection, the three phase wires are connected according to the delta configuration diagram. 4.6 DOL (Direct-On-Line) starter is connected to the motor terminals, including overload protection and control circuits. 4.7 The motor is tested, and rotation direction is verified and adjusted if necessary to ensure correct functionality.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Manuals	1.1 Manufacturer's specification manual 1.2 Repair manual 1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Manual of instruction
2. Drawings	2.1 Technical drawing 2.2 Sketch 2.3 Blue print 2.4 Electrical drawings 2.5 Connection diagram
3. Symbols	3.1 Drawing symbols 3.2 Connection symbol 3.3 Load symbol 3.4 Socket symbol 3.5 Main switch symbol 3.6 Supply symbol 3.7 Danger symbol 3.8 Switch board symbol 3.9 Conduit symbol 3.10 Starter symbols 3.11 Protective device symbol 3.12 Motor symbol
4. Controlling and protective Devices	4.1 Manual Switch 4.2 Push-Button Switch 4.3 Direct On Line (DOL) Starter 4.4 Reversing Starter 4.5 Timers 4.6 Control Relays 4.7 Overload Relays 4.8 Variable Frequency Drives (VFDs)
5. Protective devices	5.1 Thermal Overload Relay 5.2 Fuses 5.3 Miniature Circuit Breaker (MCB) 5.4 Molded Case Circuit Breaker (MCCB)

	5.5 Capacitors (Start and Run Capacitors) 5.6 Phase Loss Protection 5.7 Over/Under Voltage Protection 5.8 Ground Fault Protection 5.9 Temperature Sensors
6. Motor	6.1 Single phase induction motor 6.2 Single phase Pump Motor 6.3 Three phase induction motor

Curricular Content Guide

1. Underpinning Knowledge	1.1. Types of motors (single-phase, three-phase) and their applications. 1.2. Motor control circuits and starter systems (DOL, Star-Delta). 1.3. Electrical symbols, diagrams, and technical drawings related to motor connections. 1.4. Relevant electrical standards and safety regulations. 1.5. Protective devices for motors (e.g., fuses, circuit breakers, thermal overload). 1.6. Principles of motor operation, including voltage, current, and power specifications. 1.7. Grounding and insulation requirements for motor installations. 1.8. Troubleshooting motor connections and common motor faults.
2. Underpinning Skills	2.1. Collecting manuals and diagrams required for motor connection 2.2. Select appropriate controlling and protective devices based on motor specifications. 2.3. Interpret electrical drawings and symbols accurately for motor installation. 2.4. Install controlling and protective devices as per layout plans. 2.5. Connect single-phase and three-phase motors following wiring diagrams. 2.6. Strip insulation and crimp cable lugs securely for motor connections. 2.7. Test motor operation and verify proper functionality after connection. 2.8. Adjust motor rotation direction if needed. 2.9. Apply safety standards and ensure compliance during installation and testing.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in workplace 3.8. Communication with peers and seniors in workplace
4. Resource Implications	4.1. Workplace (simulated or actual)

	4.2. Materials, tools and equipment needed for the activity 4.3. Instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Ability to correctly identify and select appropriate controlling and protective devices for motor connections. 1.2 Competence in installing both single-phase and three-phase motors in accordance with specifications. 1.3 Proper interpretation and use of technical drawings, wiring diagrams, and electrical symbols. 1.4 Ensuring safe and compliant connections and adherence to regulatory standards. 1.5 Ability to test and troubleshoot motor connections effectively, ensuring correct operation and performance
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio review
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE RESIDENTIAL GENERATOR**	**Nominal Duration:** 35 Hrs**.**	**Unit Code:** SICIP-CON-EIM-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate residential generator. It specifically includes the tasks of installing and connecting Automatic Transfer Switch (ATS) and changeover switch, connecting the generator to the main electrical panel, preparing generator for operation, starting and stop generator and performing post-operation checks and maintenance.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Install and connect Automatic Transfer Switch (ATS) and changeover switch	1.1 ATS is selected based on the power rating and compatibility with the generator and electrical system. 1.2 ATS is installed in compliance with local electrical codes and safety regulations. 1.3 Connection between the generator and ATS is made according to the manufacturer's guidelines. 1.4 The ATS is tested for smooth transfer between mains power and generator power. 1.5 Grounding and safety connections are checked to ensure proper operation and electrical protection.
2. Connect the generator to the main electrical panel	2.1 **<u>Components of substation</u>** and **<u>main electrical panel</u>** are described. 2.2 Main electrical panel is inspected to confirm compatibility with the generator. 2.3 Proper cables and connections are selected to handle the load between the generator and substation. 2.4 Connections are securely made with attention to safety protocols and grounding. 2.5 Voltage levels are measured to ensure they match the substation or main panel requirements. 2.6 Testing is conducted to verify smooth power flow from the generator to the substation or panel.
3. Prepare generator for operation	3.1 Generator is placed in a well-ventilated area to avoid fume buildup. 3.2 Fuel level is checked and topped up as required, ensuring the correct fuel type is used. 3.3 Engine oil and coolant levels are inspected and filled to optimal levels. 3.4 Electrical connections, cables, and terminals are checked for wear or damage. 3.5 Safety switches and grounding connections are verified before operation.
4. Start and stop generator	4.1 Generator is started following the manufacturer's

	recommended procedures. 4.2 Warm-up time is allowed as specified for the generator model. 4.3 Load is applied gradually to prevent electrical surges and mechanical stress. 4.4 Generator is monitored for unusual noises, vibrations, or performance issues during operation. 4.5 Proper shutdown procedures are followed, allowing cool-down time before switching off.
5. Perform post-operation checks and maintenance	5.1 Generator is inspected after use to identify any wear, damage, or malfunctions. 5.2 Oil and fuel filters are checked and replaced, if necessary, after extended operation. 5.3 Cables, connectors, and load terminals are inspected and maintained to ensure they remain in good condition. 5.4 Battery (if applicable) is charged or replaced as required for next operation. 5.5 Generator is securely stored, and maintenance logs are updated as per workplace requirements.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Components of substation	1.1 Transformer 1.2 High Voltage Switchgear 1.3 Low Voltage Switchgear 1.4 Earthing and Grounding System 1.5 Metering and Protection Systems 1.6 Surge Protection Devices (SPDs)
2. Component of main electrical panel	2.1 Main Incoming Breaker (MCCB/ACB) 2.2 Busbars 2.3 Sub-Circuit Breakers (MCBs/MCCBs) 2.4 Automatic Transfer Switch (ATS) 2.5 Generator Control Circuitry 2.6 Surge Protection Devices (SPDs) 2.7 Energy Meters 2.8 Voltage Stabilizer or Automatic Voltage Regulator (AVR) 2.9 Neutral and Earthing Connection 2.10 Capacitor Banks

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and functions of generators and their components. 1.2 Operation and function of Automatic Transfer Switch (ATS) and changeover switches. 1.3 Electrical codes and safety regulations for generator

	installation. 1.4 Power ratings and voltage levels for residential generators. 1.5 Principles of load handling and power distribution. 1.6 Grounding, earthing systems, and electrical protection devices. 1.7 Maintenance procedures for fuel systems, engine oil, and cooling systems. 1.8 Troubleshooting methods for generator malfunctions.
2. Underpinning Skills	2.1 Select appropriate ATS and changeover switches based on power ratings. 2.2 Install ATS and generator connections following manufacturer guidelines. 2.3 Connect the generator to the main electrical panel safely and securely. 2.4 Inspect electrical connections, cables, and terminals for wear or damage. 2.5 Monitor the generator during operation for any unusual behavior. 2.6 Start and stop the generator according to safety protocols. 2.7 Perform post-operation checks and routine maintenance on the generator. 2.8 Test voltage levels to ensure correct power flow
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Learning manual 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Ability to install and connect generators, ATS, and changeover switches properly. 1.2 Competence in ensuring generator compatibility with the main electrical panel and load requirements.

	1.3 Accurate inspection and testing of electrical connections and power flow. 1.4 Safe operation and monitoring of the generator during use. 1.5 Consistent post-operation checks, maintenance, and record-keeping
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INSTALL AND TROUBLESHOOT SOLAR ELECTRICAL SYSTEM**	**Nominal Duration:** 35 Hrs**.**	**Unit Code:** SICIP-CON-EIM-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install and troubleshoot solar electrical system. It specifically includes the tasks of estimating electrical load of customer, identifying tools, equipment, and materials, setting solar panel, installing solar home system and accessories, diagnosing faults in solar home system unit and wiring, repairing faults in solar home system unit and wiring and cleaning and storing the tools and materials.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Estimate electrical load of customer	1.1 Customer required **<u>electrical load</u>** are estimated 1.2 Layout drawing of selected work plan is prepared 1.3 Capacity of panel, battery, inverter, charge controller and **<u>other accessories</u>** are selected as per guidance 1.4 Following the layout plan required quantity and size of cable, wire, and, other installation materials are estimated 1.5 Information on cost of **<u>equipment, accessories and materials</u>** collected from suppliers / manufacturers 1.6 Cost of equipment accessories and materials are estimated 1.7 Installation charges are estimated
2. Identify tools, equipment, and materials	2.1 **<u>Tools</u>** are selected and collected 2.2 **<u>Installation materials</u>**, and, **<u>solar electrical system components and accessories</u>** are collected 2.3 Battery is collected and tested
3. Set solar panel	3.1 Personal Protective Equipment (PPE) is used while working 3.2 Special rope, safety belts, and ladder are used for working on the roof 3.3 Frames are constructed as per panel size 3.4 Appropriate place with maximum sunlight exposure for panel setting is located 3.5 Setting of panels within frame is demonstrated between 23 to 30 degrees
4. Install solar home system and accessories	4.1 Solar home system and accessories are installed as per layout plan 4.2 Channel or conduit wiring is performed as per layout diagram 4.3 Switches and sockets on board are fixed 4.4 Connections with all related components are

	performed 4.5 Testing of solar electrical system for operation is performed
5. Diagnose faults in solar home system unit and wiring	5.1 **Physical faults** in the inverter, charger, charge controller, panel, battery, and wiring system are checked visually 5.2 **Operational faults** in the inverter and charge controller are checked by **testing instrument** 5.3 Panel is tested for appropriate functioning 5.4 Battery is checked by meter for appropriate voltage 5.5 Electrolyte of battery is checked by hydrometer 5.6 **Electrical connections** are checked throughout the wiring 5.7 Charge controller and inverter are tested
6. Repair faults in solar home system unit and wiring.	6.1 Burn components are replaced 6.2 Inactive and faulty components are replaced 6.3 Battery water is added if required 6.4 Loose Connections are repaired throughout the wiring
7. Clean and store the tools and materials	7.1 Tools and equipment are cleaned 7.2 Tools, measuring instruments, and access materials are stored as per work place standards

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Electrical Load	1.1 Light fixtures 1.2 Ceiling fans 1.3 Television 1.4 Refrigerator 1.5 Water pump 1.7 Computer
2. Other accessories	1.6 Light fixtures 1.7 Switch board 1.8 Switch 1.9 Sockets 1.10 MCB 1.11 Cables and wires
3. Equipment, accessories, and material's	1.12 Solar panel 1.13 Charge controller 1.14 Battery 1.15 Inverter 1.16 Switch and sockets 1.17 Cables / wires 1.18 Conduit 1.19 Fixing materials
4. Tools	1.20 Screw drivers 1.21 Diagonal cutting pliers 1.22 Long nose pliers

	1.23 Combination pliers 1.24 Adjustable wrenches 1.25 Socket wrench set 1.26 Hammer 1.27 Electric hand drill with bits 1.28 Sprit level 1.29 Measuring tape 1.30 Neon tester 1.31 Battery tester.
5. Installation materials	1.32 Cables 1.33 Channels 1.34 Screws 1.35 Rawl plugs 1.36 Clips 1.37 Nails 1.38 Plastic board 1.39 Conduits 1.40 Plastic connectors 1.41 Cable ties
6. Solar home system components	1.42 Panel 1.43 Inverter 1.44 Charge controller 1.45 Battery
7. Solar home system accessories	1.46 Light fixtures 1.47 Switches 1.48 Sockets 1.49 Junction boxes
8. Physical faults	1.50 Burn components by high temperature 1.51 Damaged by insect 1.52 Disconnection due to vibration 1.53 Loose connection
9. Operational faults	1.54 Components are inactive by aging 1.55 Components are inactive by transient effects 1.56 Components are inactive due to manufacturing defects 1.57 Components are inactive due to overload
10. Testing instruments	1.58 Volt meter (Analogue / Digital) 1.59 Ammeter (Analogue / Digital) 1.60 Multi meter (Analogue / Digital) 1.61 Hydro meter
11. Electrical connections	11.1 Terminal connections of switches, sockets, and light fixtures. 11.2 Terminal connection of panel 11.3 Terminal connection of charge controller 11.4 Terminal connection of inverter. 11.5 Terminal connection of battery.

Curricular Content Guide

1. Underpinning Knowledge	1.1. Draw layout & compute load 1.2. Selection of panel, battery, inverter, charge controller and other accessories. 1.3. Personal Protective Equipment (PPE) 1.4. Technique connects solar home system 1.5. Technique of testing solar home system operation 1.6. Diagnose faults in solar home system unit and wiring.
2. Underpinning Skills	1.1 Interpreting required electrical load. 1.2 Preparing layout drawing for selected work. 1.3 Constructing frame as per panel size. 1.4 Installing solar home system and accessories as per layout plan. 1.5 Performing solar home system testing for operation. 1.6 Testing charge controller and inverter. 1.7 Replacing burn component. 1.8 Repairing loose connection through wiring.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Learning manual 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Prepared layout drawing for selected work. 1.2 Constructed frame as per panel size. 1.3 Installed solar electrical system and accessories as per layout plan. 1.4 Performed solar home system testing for operation. 1.5 Tested charge controller and inverter. 1.6 Replaced burn component
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test

	2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Electrical Installation & Maintenance (Construction)
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	1
2.	Multimedia projector & screen	1
3.	Channel wiring practice board/booth	25
4.	Concealed wiring practice booth/wall	5
5.	Main Distribution Board with busbar, CB and other accessories (Training model)	5
6.	Sub Distribution Board with busbar, CB and other accessories (Training model)	5
7.	DOL Starter	10
8.	Generator (2 to 5 KW)	1
9.	ATS	3
10.	Changeover switch	3
11.	Solar power system (Solar panel, battery, charge controller and load)	5
12.	Electric drill machine	10
13.	Electric grinding machine	10
14.	Energy Meter (1 Phase)	10
15.	Energy Meter (3 Phase)	5
16.	Earth tester	2
17.	Megger	2
18.	Phase sequence meter	1
19.	Single phase induction motor	5

20.	Three phase induction motor	3
21.	Ceiling fan	5
22.	Pump Motor	2
23.	Bench Vice	5

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Electrical Installation and Maintenance (EIM) the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.